

It is now accepted that DNA functions in the transmission of genetic information from a parent cell to a daughter cell. Support for this concept most likely depended on demonstrating that which of the following elements were passed from parent cells to daughter cells?

I. Phosphorus

II. Sulfur

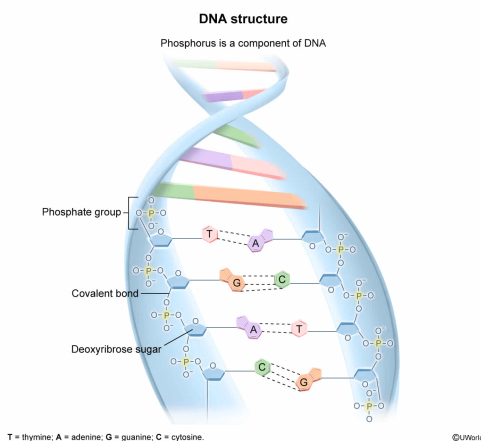
III. Carbon

- ✓ ☒ A. I only [35%]
☐ B. II only [4%]
☐ C. I and III only [55%]
☐ D. II and III only [4%]

Correct

35% answered correctly

Explanation:



DNA is a polymer consisting of **deoxyribonucleotide monomers**, typically in a double-stranded format (single-stranded DNA occurs in some viruses and during DNA replication). Paired DNA strands are aligned in an **antiparallel direction** and wrap around one another, forming a double helix. The nucleotide building blocks of DNA are composed of a nucleoside molecule attached to one or more phosphate groups via a phosphoester bond. Nucleosides consist of a pentose (five-carbon) sugar linked to a nitrogenous base.

DNA consists of an **exterior "backbone"** and an interior "ladder." The exterior backbone of DNA is made of alternating deoxyribose (five-carbon/pentose) sugars and phosphate groups. The 3'-hydroxy group of the deoxyribose from a deoxyribonucleotide monomer binds via **phosphodiester linkage** to the 5'-phosphate group of the adjacent deoxyribonucleotide monomer.

The interior ladder of DNA is composed of complementary **nitrogenous bases** that are attached to the deoxyribose sugars of both strands. DNA contains four nitrogenous bases: the purines adenine (A) and guanine (G), and the pyrimidines cytosine (C) and thymine (T). These nitrogenous bases stay in this ladder formation via **hydrogen bonds**.

This question states that it is now accepted that DNA functions in genetic information transmission from a parent cell to a daughter cell. To provide support for this idea, scientists likely demonstrated that elements unique to DNA (ie, instead of elements unique to another **biomolecule**) were passed from parent cells to daughter cells. One of these elements is **phosphorus**, a component of the phosphate groups making up DNA (**Number I**).

(Number II) Sulfur is an element found in [proteins](#), *not* nucleic acids.

(Number III) Although carbon is a component of DNA, it is also a component of the other three biomolecules: proteins, lipids, and [carbohydrates](#). Therefore, demonstrating that carbon is passed from parent cells to daughter cells would *not* provide definitive support that DNA, rather than another biomolecule, functions in the transmission of genetic information.

Educational objective:

DNA is a double-stranded polymer made of nucleotides, consisting of a backbone made of deoxyribose sugars and phosphate groups that alternate with one another. From this backbone project four nitrogenous bases (adenine, thymine, guanine, cytosine) that bind to the complementary nitrogenous base in the second, antiparallel DNA strand.

Time spent:0 sec

Subject:
Biology

QID:403796

System:
Molecular Biology